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Electron Capture vs Transfer Dissociation for Site Determination of Tryptic Peptide Tyrosine Sulfation: Direct Detection of Fibrinogen Sulfation Sites and Identification of Novel Isobaric Interferences

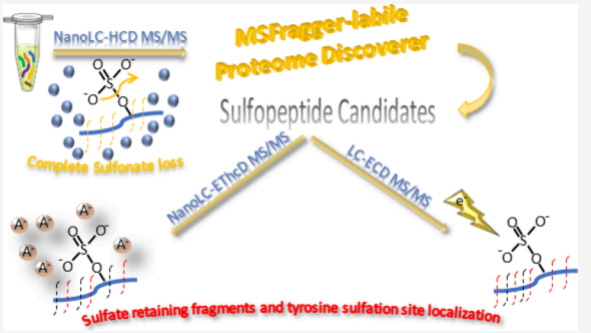
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Cite This: *J. Proteome Res.* 2024, 23, 2386−2396

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ABSTRACT: Tyrosine sulfation, an understudied but crucial post-translational modification, cannot be directly detected in conventional nanoflow liquid chromatography-tandem mass spectrometry (nanoLC-MS/MS) due to the extreme sulfate lability. Here, we report the detection of sulfate-retaining fragments from LC-electron capture dissociation (ECD) and nanoLC-electron transfer higher energy collision dissociation (EThcD). Sulfopeptide candidates were identified by Proteome Discoverer and MSFragger analysis of nanoLC-HCD MS/MS data and added to inclusion lists for LC-ECD or nanoLC-EThcD MS/MS. When this approach failed, targeted LC-ECD with fixed m/z isolation windows was performed. For the plasma protein fibrinogen, the known pyroglutamylated sulfopeptide QFPTDYDEGQDDRPK from the beta chain N-terminus was identified despite a complete lack of sulfate-containing fragment ions. The peptide QVGVEHHVEIEYD from the gamma-B chain C-terminus was also identified as sulfated or phosphorylated. This sulfopeptide is not annotated in Uniprot but was previously reported. MSFragger further identified a cysteine-containing peptide from the middle of the gamma chain as sulfated and deamidated. NanoLC-EThcD and LC-ECD MS/MS confirmed the two former sulfopeptides via sulfate-retaining fragment ions, whereas an unexpected fragmentation pattern was observed for the third sulfopeptide candidate. Manual interpretation of the LC-ECD spectrum revealed two additional isobaric identifications: a trisulfide-linked cysteinyl-glycine or a carbamidomethyl-dithiothreiotol covalent adduct. Synthesis of such adducts confirmed the latter identity.



KEYWORDS: *tyrosine sulfation, direct PTM determination, site localization, mass spectrometry, post-translational modifications, proteomics, electron capture dissociation, electron transfer dissociation, cysteinyl-glycine*

INTRODUCTION

Tyrosine-*O*-sulfation in eukaryotes is catalyzed by tyrosyl protein sulfotransferases TPST-1 and TPST-2, which transfer an SO_3 group to the hydroxyl group of a tyrosine (Tyr) residue with 3′-phosphoadenosine 5′-phosphosulfate (PAPS) as a source of sulfate.[1] TPSTs are membrane-bound enzymes located in the trans-Golgi network. Thus, sulfation occurs as proteins are translocalized to the cell membrane or secreted.[2] Although Tyr sulfation is anticipated to occur in about 1% of eukaryotic proteins, it is an understudied post-translational modification (PTM) with only a few sulfoproteins identified.[3] Recent advances in enrichment strategies, data acquisition strategies, and bioinformatics utilities allowed the first sulfoproteomic analyses of Golgi resident proteins[4] as well as secreted proteins from HEK-293 cells.[5] Tyr sulfation can play a critical role as a part of the "interactome" (i.e., the interaction of a protein with other biomolecules) by regulating extracellular protein−protein interactions to, e.g., harmonize hemostasis, and play roles in leukocyte adhesion, chemokine signaling, as well as inflammatory response.[6−8] In addition, there is evidence for cross talk with other PTMs, for example, colocalization of sulfotyrosine and *O*-glycosylation in crucially functional domains,[9] and sulfo- and phospho-tyrosine in Golgi proteins.[4]

Despite the growing interest in mapping tyrosine sulfation owing to its biological significance, a limited number of analytical methods have been reported for identification of this PTM. In tandem mass spectrometry (MS/MS), the gold standard for PTM identification/localization, sulfopeptide characterization is highly challenging because of the extreme gas-phase lability and low ionization efficiency of sulfopeptides

Received: February 2, 2024
Revised: April 30, 2024
Accepted: June 7, 2024
Published: June 20, 2024



ACS Publications

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in positive ion mode.[10] Sulfated (+79.9568 Da) and phosphorylated (+79.9663 Da) tyrosines can also be misassigned because of their isobaric nature; however, high-resolution mass spectrometers can differentiate between these two PTMs at the MS^1 level. Furthermore, PTM-containing fragments are observed for phosphopeptides but not sulfopeptides in collision-induced dissociation (CID)/higher energy collision dissociation (HCD or beam-type CID), which shows predominant neutral SO_3 loss.[5,11,12] Complete SO_3 elimination can also occur upon electron capture/transfer dissociation (ECD/ETD)[12−14] depending on sulfopeptide sequence/basicity and charge state.[15] Sulfation retention for highly basic sulfopeptides has been observed, most likely due to the formation of a salt bridge between arginine side chains and the acidic sulfo-PTM.[12,14] A different approach to localize sulfation sites in positive ion mode employs the instability of the sulfate PTM in deductively based identification methods. These methods involve unmodified tyrosine acetylation or bromination prior to MS/MS analysis.[16,17] In these approaches, tyrosine residues detected without modification are attributed to the loss of sulfate upon CID. The latter techniques can be effective; however, they do necessitate a quantitative reaction of unmodified tyrosines to avoid false positives and also require additional upfront sample processing.

Alternatively, employing MS/MS analysis in negative ion mode offers a more direct approach for the detection and localization of tyrosine sulfation. This approach is based on the enhanced stability of sulfopetides as gas-phase anions. However, CID of peptide anions leads to preferential SO_3 neutral loss. Although this fragmentation pathway confirms the presence of sulfation, the absence of peptide backbone fragments poses a significant limitation.[18,19] With the need for improved MS/MS characterization, alternative activation methods such as metastable atom-activated dissociation[20] and negative ion electron capture dissociation[21,22] show promise for tyrosine sulfation mapping. Both techniques offer extensive peptide sequence coverage without significant SO_3 losses. However, these two methods require relatively long activation times or extensive spectral averaging for optimal results, which restrict their compatibility with online liquid chromatography (LC)-MS methods. Alternatively, negative ion free radical initiated peptide sequencing (nFRIPS) has been shown to also retain sulfation in fragment ions;[23] however, this technique requires extensive derivatization and has not yet been demonstrated in combination with LC. Ultraviolet photodissociation (UVPD) at 193 nm is a fast activation method, which has been applied in combination with LC separation for the analysis of sulfated peptides in negative ion mode showing diagnostic neutral SO_3 loss. However, the sulfation site could be confidently assigned because of the observation of a series of sulfate-retaining *a*/*x*-type product ions.[24] One drawback of nFRIPS and UVPD is that the resulting MS/MS spectra are complex with a variety of fragment ion types, rendering spectral annotation difficult. Electron detachment dissociation (EDD)[25] has also been coupled with LC separation;[26] however, despite showing partial sulfate retention in fragment ions,[22] EDD is an inefficient MS/MS method showing abundant uninformative neutral losses. The latter drawback also affects negative ion electron transfer dissociation,[22] the ion−ion reaction analogue of EDD.

Despite the absence of sulfate-retaining fragments in conventional CID/HCD-based MS/MS analysis, sulfopeptide candidates can be identified via characteristic SO_3 loss upon low energy HCD[5] or via MSFragger[27]-based precursor ion accurate mass shift analysis as compared with an identified, unmodified tyrosine-containing peptide sequence.[4] However, these CID/HCD-based approaches only indirectly assign tyrosine sulfation and cannot identify the sulfation site if more than one tyrosine residue is present. The aim of this study is to directly detect and localize tyrosine sulfation sites in a proteomics workflow. To meet that objective, we combined bottom-up nanoflow LC (nanoLC)-HCD with conventional + "MSFragger-Labile"[28] analysis, nanoLC-electron transfer with supplemental HCD (EThcD)[29] MS/MS, and LC-ECD MS/MS for the characterization of bovine fibrinogen sulfopeptide candidates both from purified protein and from plasma.

EXPERIMENTAL PROCEDURES

Trypsin Digestion

Bovine fibrinogen and bovine plasma (Sigma, St. Louis, MO, USA) stock solutions were prepared in 50 mM ammonium bicarbonate (Sigma). For trypsin digestion, aliquots were diluted to 1 mg/mL protein concentration with 100 mM urea (Sigma) in 50 mM ammonium bicarbonate. Samples were reduced with 200 μM dithiothreitol (DTT; Sigma) at 37 °C for 30 min and then alkylated with 400 μM iodoacetamide (IAA, Sigma) for another 30 min at 37 °C in the dark. Sequencing grade modified trypsin (Promega, Madison, WI, USA) was added in a 1:50 enzyme/protein ratio, and digestion occurred at 37 °C overnight. Formic acid (Fisher Scientific, Fair Lawn, NJ, USA) was then added to the digested protein mixture to a final concentration of 0.5% to stop the trypsin activity. Tryptic peptides were desalted using Pierce C18 Spin Columns (Thermo Scientific, USA, product number 89873) before LC-MS/MS analysis.

For tris(2-carboxyethyl)phosphine (TCEP, Sigma) reduction, a similar protocol for trypsin digestion was followed, adding TCEP to a final concentration of 5 mM with incubation for 20 min at room temperature followed by addition of IAA at the same concentration with incubation for 15 min in the dark at room temperature. Another protocol of incubating the protein first with equal volume of 8 M urea for 1 h at 37 °C was adopted to enhance protein denaturation and unfolding.[30] TCEP was then added as in the previous protocol, and IAA was added to a final concentration of 10 mM.

NanoLC-HCD MS/MS and Neutral Loss Triggered-HCD MS/MS

LC-MS grade solvents and mobile phase additives were purchased from Fisher Scientific. NanoLC separation was carried out on an UltiMate 3000 RSLCnano System (Thermo Fisher Scientific, Dionex Softron GmbH, Germering, Germany) with HCD MS/MS data-dependent acquisition on an Orbitrap Fusion Lumos Tribrid mass spectrometer (Thermo Fisher Scientific, San Jose, CA, USA). Desalted tryptic peptides were injected on a nanoViper precolumn (75 μm i.d. × 2 cm Acclaim PepMap 100 C18, 3 μm particle size, 100 Å pore size, Thermo Fisher). Peptide separation was then achieved on a 50 cm column (75 μm i.d. × 50 cm Acclaim PepMap 100 C18, 3 μm particle size, 100 Å pore size, Thermo Fisher) set at 46 °C with gradient elution at a 300 nL min^{-1} flow rate. Mobile phase B (acetonitrile/water 80:20% v/v with 0.1% formic acid) was held at 2% for 5 min and then increased to 40% over 40 min followed by a ramp to 95% over 5 min. The flow was then held at 95% B for 5 min and 2% B for another 5 min for

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washing and re-equilibration. Mobile phase A was water/acetonitrile 98:2% v/v with 0.1% formic acid.

MS^1 scans were acquired in the Orbitrap over the m/z range 350 to 1700 with 120k resolution at 200 m/z, maximum injection time = 50 ms, and normalized automatic gain control (AGC) = 100%. Dynamic exclusion was applied after one scan for 20 s at a 10 ppm mass tolerance. Ions with a minimum intensity of 5.0e4 were selected for HCD MS/MS fragmentation with detection in the Orbitrap with 60k resolution at 200 m/z, maximum injection time = 118 ms, and normalized AGC target = 100%. HCD normalized collision energy (NCE) was set to 32%, and internal calibration was selected.

Neutral loss triggered (NLT)-HCD MS/MS was performed with the same MS^1 parameters. HCD MS/MS with NCE set to 10% was acquired in the Orbitrap with 30k resolution at 200 m/z. Targeted acquisition was triggered for SO_3 (79.9568 Da) with 25 ppm mass tolerance. NLT scans were detected in the Orbitrap (30k resolution at 200 m/z) with HCD set to 30% NCE, maximum injection time of 54 ms, and normalized AGC target = 20%.

Targeted NanoLC-ETD with Supplemental HCD (EThcD) MS/MS

Sulfopeptide candidates from HCD data were added to a targeted inclusion list to perform EThcD at NCE 20, 30 and 40% with detection in the Orbitrap using 30k resolution at 200 m/z. ETD spectra were acquired using the calibrated charge-dependent ETD parameters, a maximum injection time of 54 ms, and a normalized AGC target = 100%.

LC-ECD MS/MS

Sulfopeptide candidates were added to an inclusion list or time-segmented method for LC-ECD MS/MS with an Agilent 1290 Infinity LC system (Santa Clara, CA, USA) coupled to a 7 T SolariX Fourier transform ion cyclotron resonance (FT-ICR) mass spectrometer (Bruker Daltonics, Billerica, MA, USA) in positive ion mode. This instrument was recently upgraded with the ParaCell,[31] designed for exceptional spectral resolution and higher sensitivity. ECD was performed with an indirectly heated hollow dispenser cathode[32] operating at 1.5 A.

Tryptic peptides were separated via Agilent InfinityLab Poroshell 120 EC-C18 columns, either 4.6 × 100 mm, 2.7 μm, or 2.1 × 100 mm, 2.7 μm, at 40 °C. Mobile phase A was 0.1% formic acid in water, whereas mobile phase B was 0.1% formic acid in acetonitrile. The entire LC run was 47.1 min at a flow rate of 250 or 50 μL min^{-1} during the gradient and equilibration portions of the run for the 4.6 and the 2.1 mm column, respectively. A higher flow rate, 500 μL min^{-1}, was used during the wash for both columns. Data presented from quadrupole isolation at a fixed m/z value during the entire run or from time segmented methods were acquired with the 4.6 mm column. For further LC-ECD MS/MS experiments, the 2.1 mm column at a lower flow rate resulted in improved sensitivity. The nebulizer gas pressure, drying gas flow rate, and drying gas temperature were set to 3 bar, 10 L/min, and 250 °C, respectively.

Data-dependent LC-ECD-MS/MS was performed with MS/MS boost, and precursor masses of sulfopeptide candidates were added to the preferred list for fragmentation. Accumulation times for MS and MS/MS scans were 0.1 and 0.5 s, respectively. ECD (0.1 s irradiation time, −0.3 V ECD bias, 20 V ECD lens) was performed in the FT-ICR cell. Single scan data were acquired with a 256 Kword acquisition size. LC-ECD MS/MS was also performed either with a fixed quadrupole isolation window for the entire LC run or with time segmented methods in which quadrupole isolation at fixed m/z values was performed at defined time intervals during the LC elution. In these methods, the external accumulation time was 1 s.

Synthesis of Trisulfide-Linked Peptide Pair

Synthetic peptide QQVYCEIQK (Genscript Corp, Piscataway, NJ) and cysteinyl glycine (CG) peptide (Sigma), 50 mg each, were solubilized together in water/formic acid (99.5%) 50:50% v/v to a final concentration of 100 mg/mL peptide and incubated overnight to induce disulfide bond formation. A published protocol[33] for cystine trisulfide synthesis was adopted with minor adjustments. Briefly, 2 mol equiv of cold peracetic acid was added to the prepared disulfide-linked peptide pair solution followed by stirring for 2 h on ice. The formed *S*-monoxide species was then precipitated with diethyl ether by adding ∼3-fold the reaction volume. The precipitate, sticking to the round-bottom flask, was washed with ether two to three times and coarsely dried by blasting with nitrogen gas.

The crude *S*-monoxide product (white, sticky solid) was then immediately dissolved in 1 mL water/formic acid (99.5%) 50:50% v/v. In a separate two-necked flask, an excess of 2 mol equiv of anhydrous solid sodium hydrosulfide (Thermo Fisher Scientific) was placed along with a stir bar. This flask was then sealed with one rubber stopper and placed under an argon atmosphere before the second neck was sealed with a balloon under atmospheric/low argon pressure. The peptide solution was transferred via a cannula to the sodium hydrosulfide, initiating the reaction, and allowed to stir for 1 h. After 1 h, excess gas pressure was vented into a hydrogen sulfide neutralization slurry.[34] The generated trisulfide-linked peptide pair was precipitated (white solid), washed with ether, and dried. The precipitate was solubilized in water for direct infusion electrospray ionization (ESI)-MS/MS analysis

Data Analysis

Proteome Discoverer 2.2 (Thermo Fisher Scientific, San Jose, CA, USA) was used for analyzing the nanoLC-HCD MS/MS and NLT-HCD MS/MS data. Raw spectra were searched against bovine fibrinogen or *Bos taurus* canonical and isoform FASTA libraries downloaded from UniProtKB (Release 2023_04) for fibrinogen and plasma samples, respectively. Mass tolerances of 5 ppm and 0.02 Da were set for the precursor and fragment ions, respectively. The search parameters were adjusted allowing two missed tryptic cleavages and variable modifications including M-oxidation, N-terminal acetylation, Y-sulfation, and S-, T-, and Y-phosphorylation. C-carbamidomethylation was set as a fixed modification.

LC-HCD MS/MS data were also analyzed with MSFragger (version 3.7). MSConvert (version 3.0.22331) from ProteoWizard[35] was used to convert raw files to the mzML centroided format used in subsequent searches. All searches were conducted using FragPipe (version 19.1) and Philosopher (version 5.0). The labile-phospho workflow template was applied and was also customized to search for sulfation. The database search was performed against a reviewed bovine proteome with added decoys and common contaminants, downloaded on October 17, 2023, containing 12,290 total entries, including contaminants. MSFragger searches utilized built-in mass calibration, a strict-trypsin enzymatic search with allowance for two missed cleavages, along with precursor and

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fragment tolerances of 20 ppm. The search also included fixed modifications, including cysteine carbamidomethylation, as well as variable modifications, including methionine oxidation, protein N-terminal acetylation, and tyrosine phosphorylation or sulfation. Validation was performed using PTMProphet for the final localization of all modifications employing a static 20 ppm fragment tolerance. ProteinProphet was utilized for protein inference with a false discovery rate filter at 1%.

ECD MS/MS spectra were internally calibrated using fragment ions assigned with high confidence. For LC-ECD MS/MS data annotation, an in-house custom method was developed with the Bruker DataAnalysis software (version 5.2). This method compares monoisotopic theoretical fragment ion m/z values and the corresponding charge states with deisotoped experimental data and incorporates the advantage of specifying a mass tolerance for this comparison. Deisotoping to determine experimental monoisotopic m/z values and charge states was performed with the Bruker SNAP algorithm. The method reads the theoretical m/z values and charge states from a comma separated value (CSV) text file and searches for matches between this mass list and the list of observed monoisotopic m/z values and charge states determined by SNAP within the provided mass tolerance and the correct charge state. The method also generates annotation objects in the corresponding MS/MS spectrum. Each annotation object encompasses information about the matched theoretical fragment ion and the associated parts per million error for the assignment. Here, 10 ppm error was used. Theoretical fragment ion mass lists were generated with Protein Prospector (version 6.4.9). Peptide sequence coverage and degree of sulfate retention were displayed by using an in-house script.

RESULTS AND DISCUSSION

Proteome Discoverer Analysis of Fibrinogen NanoLC-HCD MS/MS Data

NanoLC-HCD MS/MS data of tryptically digested bovine fibrinogen obtained on the Orbitrap Fusion Lumos were first analyzed with Proteome Discoverer using both MS Amanda[36] and Sequest[37]-type searches with the bovine fibrinogen FASTA.

Proteome Discoverer identified two peptides from bovine fibrinogen as sulfated or phosphorylated (Figure 1a). One of these peptides is a known sulfopeptide, •QFPTDY■DEGQDDRPK (• = pyroglutamic acid; Y■ = sulfotyrosine) from the N-terminus of the fibrinogen beta chain (residues 1−15).[24] Upon inspection of a corresponding HCD spectrum (Figure 1b), no sulfated fragment ions were observed, as expected. However, high sequence coverage from 21 observed *b*- and *y*-type fragments provided confident identification of this sulfopeptide (MS Amanda score = 232.35). By contrast, this peptide was identified as phosphorylated by Sequest (Xcorr score = 2.74), which did not match the observed complete PTM loss (Figure 1b). The second peptide, QVGVEHHVEIEY■D, from the C-terminus of the fibrinogen gamma-B chain (residues 432−444) was also identified as sulfated by MS Amanda (score = 357.36) and as phosphorylated by Sequest (XCorr score = 2.26). Similar to the known sulfopeptide, manual spectral interpretation showed a complete neutral loss of 79.9568 from the peptide fragments. This neutral loss differentiates tyrosine sulfation from tyrosine phosphorylation because only partial loss of HPO_3 (79.9663) is observed from phosphotyrosine. In addition, mostly H_3PO_4 loss (97.98 Da) is observed from serine- and threonine-phosphorylated peptides.[4] Thus, the sulfate modification is confirmed (Figure 1c). This peptide also showed high sequence coverage (24 *a*-, *b*-, and *y*-type fragments). However, because of the complete SO_3 elimination, direct detection of tyrosine sulfation is not achieved for either peptide. The latter sulfation site is not annotated in UniProt, but Tyr 443 sulfation has been previously reported from indirect evidence.[38]

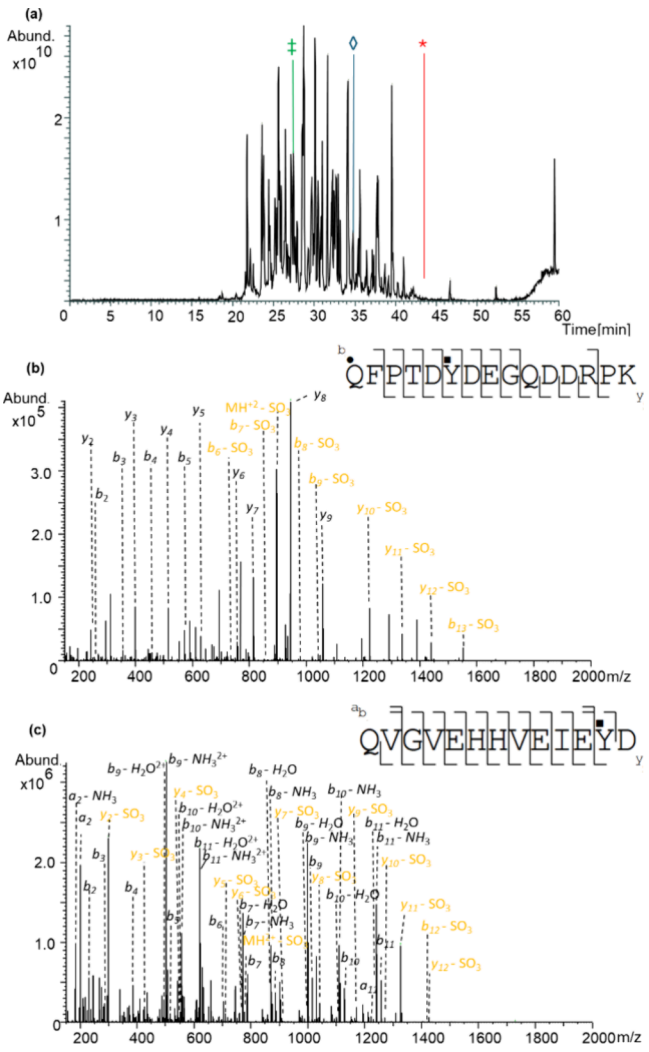


Figure 1. NanoLC-HCD MS/MS of tryptically digested bovine fibrinogen on an Orbitrap Fusion Lumos. Total ion chromatogram (a) and HCD MS/MS spectra of two identified doubly protonated sulfopeptides: •QFPTDY■DEGQDDRPK (• = pyroglutamic acid; Y■ = sulfotyrosine; b) and QVGVEHHVEIEY■D (c), eluting at 34.6 min (blue diamond) and 27.3 min (green double dagger), respectively. No sulfate-retaining fragments are observed, as expected.

The same two sulfopeptides from the termini of the fibrinogen beta and gamma-B chains were also successfully identified with the same nanoLC-HCD MS/MS method directly from bovine plasma, a significantly more complex mixture (Figure S1). In addition, NLT-HCD MS/MS with low collision energy[5] identified both sulfopeptides, further confirming these sulfation sites (Figure S2). However, because sulfopeptides typically have low abundances due to low ionization efficiencies, targeted sulfopeptide enrichment is necessary for sulfopeptide identification from less abundant proteins.[4,5]

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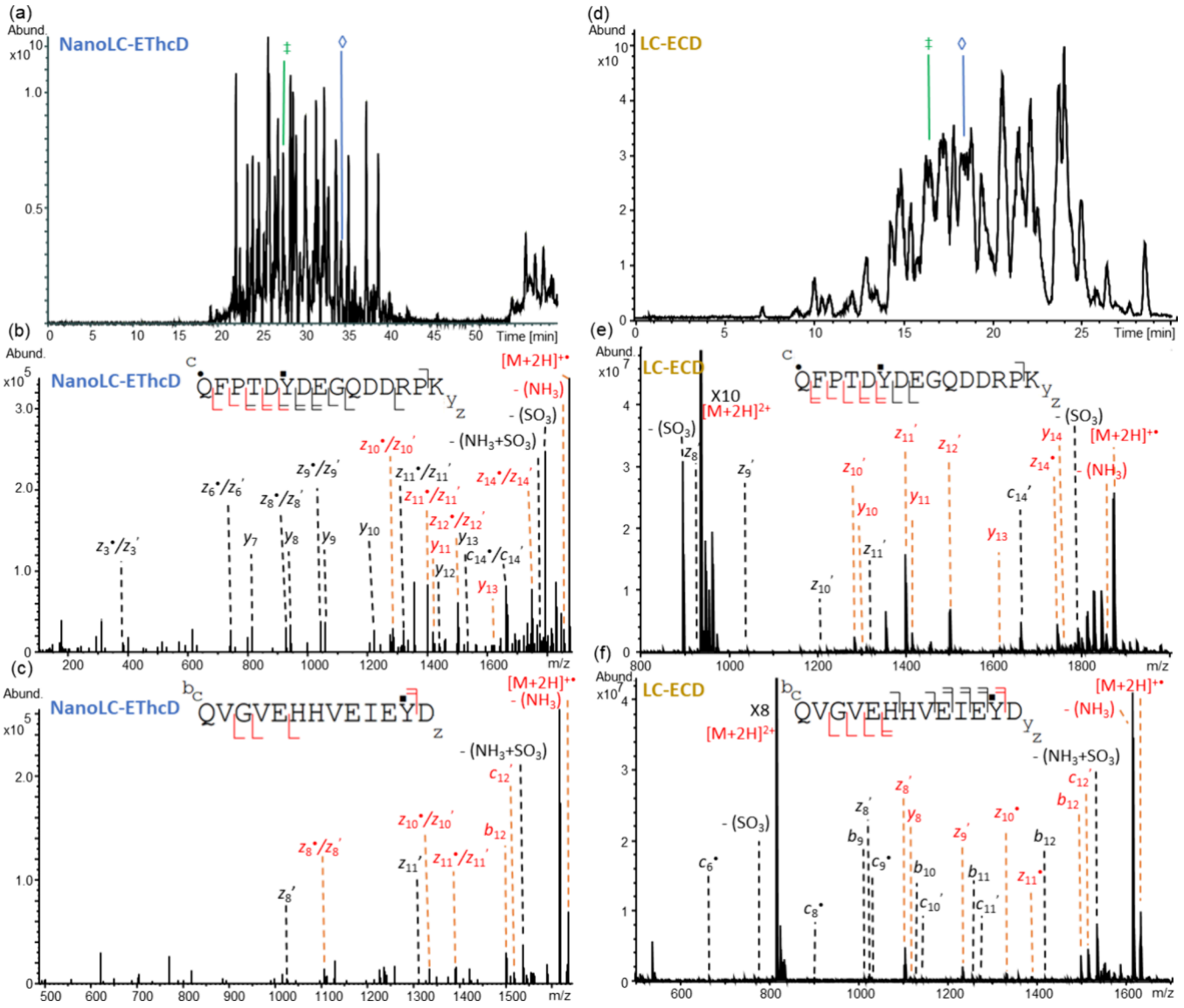


Figure 2. NanoLC-EThcD (a–c) and LC-ECD (d–f) MS/MS of tryptically digested fibrinogen showing the gamma-B (green double dagger) and beta chain (blue diamond) peptides, respectively. ECD spectra of both sulfopeptides (e, f) showed a higher number of sulfate-retaining fragment ions (in red) than EThcD spectra of the same peptides (b, c) at an optimum NCE.

NanoLC-EThcD MS/MS of Fibrinogen Tryptic Peptides at Varying Supplemental Collision Energies

With the goal of generating sulfate-containing fragment ions, an alternative MS/MS method, EThcD, was employed along with a targeted inclusion list containing the two doubly protonated sulfopeptides shown in Figure 1. ETD is recognized for providing peptide backbone cleavage while predominantly maintaining labile modifications.[39]

However, ETD typically does not perform well for doubly charged precursor ions.[40] The ETD efficiency for such low charge state precursors can be enhanced with the introduction of supplemental HCD activation.[29] Supplemental NCEs at 20, 30, and 40% were employed (Figure S3). EThcD generated more sulfate retaining fragment ions upon increasing the supplemental activation energy for the •QFPTDY■DEGQDDRPK sulfopeptide, with the highest sequence coverage obtained at NCE 40 (Figure 2b). However, the loss of SO_3 from several fragment ions is notable. At lower NCE, desulfation was less apparent but at the expense of reduced sequence coverage. In addition, at lower NCE, significant intact and desulfated 2+ precursor ion signal remained.

For the QVGVEHHVEIEY■D sulfopeptide, the same trend of enhanced fragmentation was observed upon increasing the HCD supplemental energy from 20 to 30 % (Figure S3 and Figure 2c). Similar to the beta chain sulfopeptide, no desulfation was observed at the lower NCE. At 40% NCE, no precursor ion remained, and there was no further increase in sequence coverage (Figure S3).

LC-ECD MS/MS for Improved Sulfate Retention

Although ETD and ECD generate the same fragment ion types, they differ in terms of the recombination energy.[41] When ECD is performed in an ICR cell at ultrahigh vacuum, there is also a significant difference in pressure during activation compared with ETD in an ion trap (~10^{-3} mbar), further differentiating the precursor ion internal energy. Thus, higher fragmentation efficiency can be achieved with ECD at low charge states.[42] In that context, LC-ECD MS/MS was

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performed on a SolariX FT-ICR instrument with the two fibrinogen sulfopeptides added to the preferred list for fragmentation (Figure 2d). This approach allowed these relatively low abundance peptides to be selected for ECD. Both intact and desulfated precursor ions were observed in bovine fibrinogen digests. Upon ECD, more sulfate-retaining fragment ions were detected (Figure 2e,f) compared with optimized EThcD (Figure 2b,c). ECD and EThcD both provided direct evidence of sulfation at tyrosine 443 in the fibrinogen gamma-B chain, which was not annotated in Uniprot. The same LC-ECD MS/MS method was also applied to the digested plasma. The two doubly charged fibrinogen peptides were observed and subjected to ECD. However, the spectral quality was not as high (Figure S4).

MSFragger Search for Additional Sulfopeptide Candidates

Based on our previous work showing that the open precursor ion search tool MSFragger is superior to conventional database search for identifying sulfopeptides,[4] we employed the recently introduced MSFragger "labile mode"[28] to further interrogate the nanoLC-HCD MS/MS data (Figure 1a and Figure S1) from the tryptically digested fibrinogen and plasma samples acquired on the Orbitrap Fusion Lumos. The MSFragger Labile Mode can filter spectra for diagnostic ions, characteristic of a specific labile modification, and also incorporates information about the remainder masses of modified peptide fragments, thus improving the confidence of spectral identification and modification localization. We utilized the already available workflow template for labile phosphorylation (mass shift of 79.9663 Da) and also customized it to search with the accurate SO_3 mass shift of 79.9568 Da. MSFragger reports identified peptide sequences, assigned modifications, and the mass shift from the base (unmodified) peptide sequence. This search resulted in additional sulfopeptide candidates, typically at low abundance.

To further evaluate whether these additional peptides are indeed sulfated, we performed additional targeted analysis, with most sulfopeptide candidates not being selected for fragmentation. However, the MSFragger-identified sulfopeptide candidate, AKQ◆FLVY■C*EIDGSGNGWTVFQK (◆ = deamidation; * = carbamidomethyl (CAM) cysteine), from the fibrinogen gamma-B chain (residues 198−219) showed sufficient abundance for further interrogation. The nanoLC-HCD MS/MS elution time for this peptide is highlighted in Figure 3a. The corresponding HCD spectrum (Figure 3b,c) was annotated based on the MSFragger-determined sequence (Figure 3b), showing confidently matched *y*-type ions with deamidation at Gln200; however, no *b*-type fragments were observed (Figure 3b). The complete lack of *b* ions was surprising, and thus, we performed a BLAST sequence similarity search, which reported the sequence variant, **Q**QFLVYCEIDGSGNGWTVFQK (as compared with the MSFragger-identified sequence (**AK**)-QFLVYCEIDGSGNGWTVFQK, which has a missed trypsin cleavage site), from the fibrinogen gamma chain (residues 200−220). The HCD spectrum was reanalyzed and found to better match this sequence variant, including several *b*-type ions (the *y* ions are the same), but without deamidation or sulfation modifications (Figure 3c). *b*- and *y*-type fragment ions were observed up to the cysteine residue in position 206 but not beyond this residue, suggesting a cysteine modification other than alkylation.

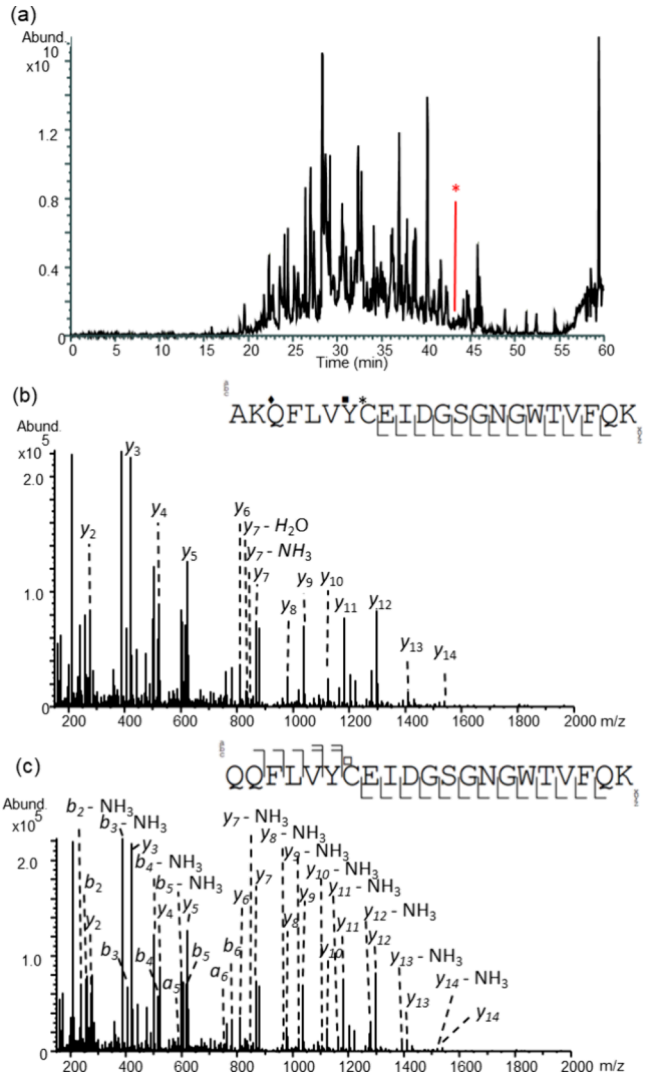


Figure 3. NanoLC-HCD MS/MS total ion chromatogram of tryptically digested plasma (a). HCD spectrum annotation of AKQ◆FLVY■C*EIDGSGNGWTVFQK (◆ = deamidation, C* = carbamidomethyl cysteine) peptide (b) suggested by MSFragger, eluting at 43.3 min (red asterisk) (a). Upon annotating the same HCD spectrum with the QQFLVYC□EIDGSGNGWTVFQK (C□ = cysteine modification) peptide, 90% sequence coverage was observed (c).

LC-ECD and nLC-EThcD MS/MS for Unraveling a Cysteine Modification

The triply charged precursor ion of the BLAST-assisted, manually identified cysteine modified QQFLVYCEIDGSGNGWTVFQK peptide from the fibrinogen gamma chain (residues 200−220) was added to the preferred list of fragmentation in LC-ECD MS/MS. However, because of its low abundance, it was still not selected for fragmentation. Therefore, a time-segmented method in which the quadrupole isolation window was set to the targeted peptide precursor ion *m*/*z* value during its elution window (Figure S5) was employed. The corresponding ECD spectrum (Figure 4a) showed confidently matched *c*- and *z*-type ions, including fragments confirming deamidation at Asn213. In addition, an unusually abundant doubly charged fragment ion, corresponding to the mass of the peptide sequence without cysteine modification ("unmodified peptide", measured monoisotopic

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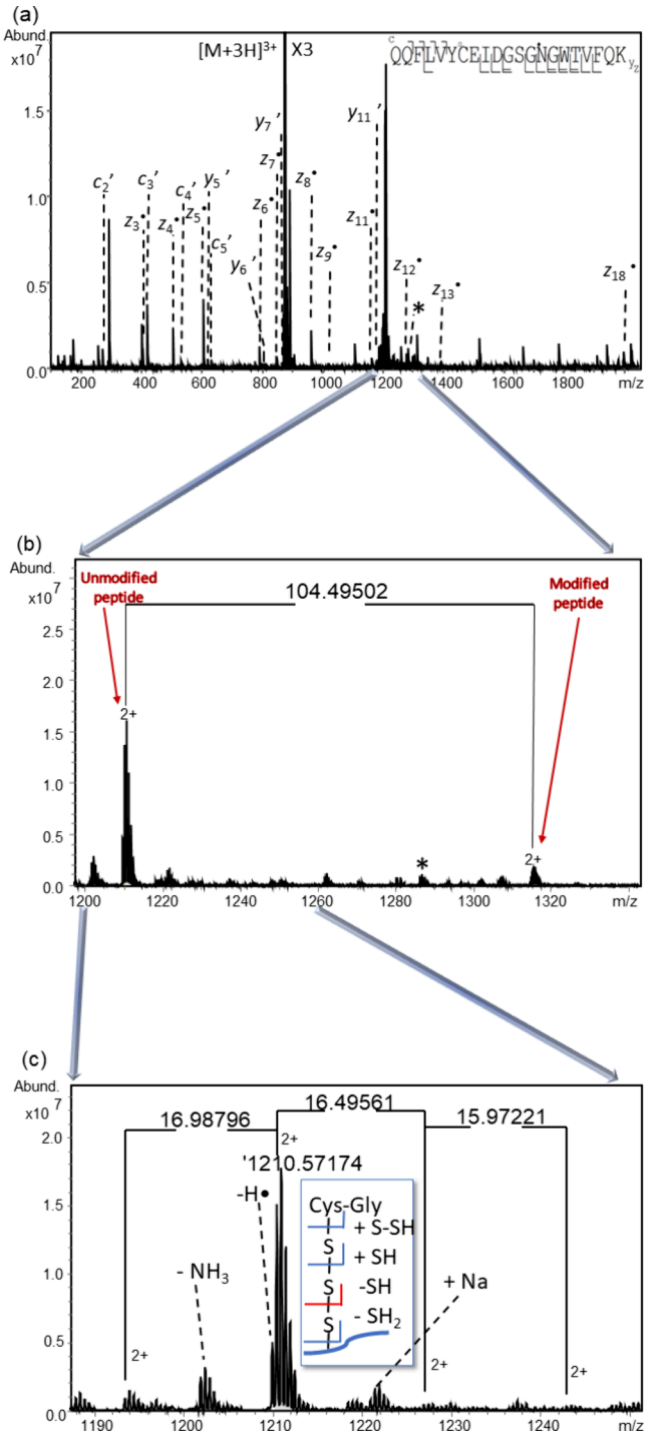


Figure 4. LC-ECD MS/MS of the modified peptide QQFLVYC□EIDGSGN◆GWTVFQK resulted in several c- and z-type fragments matching the unmodified peptide (a). A zoomed-in view of the ECD spectrum showing the mass difference between the cysteine modified and unmodified peptide (b). Characteristic satellite peaks to the abundant 2+ ion at 1210.57174 were observed (c).

mass = 2419.1289 Da), was observed (Figure 4a–c). A lower abundance species, 1 Da lighter, is also present (Figure 4c). This type of fragmentation behavior is characteristic of disulfide bond cleavage, resulting in a mixture of −S• and −SH-type fragment ions.[43] Because the even-electron −SH species is dominant, the observed neutral loss of 208.99004 Da (2 × 104.49502) (Figure 4b) must be a radical species. This mass is relatively low for a tryptic peptide. However, there are other thiol-containing molecules in plasma that could correspond to this presumed cysteine covalent adduct. An obvious candidate is glutathione, but its molecular weight is too high (307 Da). Christensen and co-workers reported disulfide-linked cysteinyl-glycine (CG) in human alpha2-antiplasmin.[44] However, the CG monoisotopic mass (178.04121 Da) is too low. On the other hand, the mass difference between a neutral loss of a CG radical (177.03431) and the observed neutral loss of 208.99004 is 31.9557, very close to an additional sulfur atom (31.9721 Da). Trisulfide bonds have been reported to be prevalent in plasma proteins;[45] thus, one hypothesis would be that the observed fibrinogen peptide, initially identified by MSFragger as a sulfopeptide, instead corresponds to a peptide with a trisulfide-linked cysteinyl-glycine adduct. The satellite peaks in Figure 4c also support this assignment, as indicated in Figure 4c inset.

We also attempted targeted mass nanoLC-EThcD MS/MS of the potential CG-adducted peptide at 30% NCE. The resulting EThcD spectrum (Figure S6) showed several matching *b*- and *y*-type fragment ions (presumably from the HCD supplementary activation); however, no fragments containing the modified cysteine were observed. Similar to the LC-ECD spectrum (Figure 4a), an unusually abundant doubly charged fragment corresponding to the unmodified peptide was observed; however, accompanying satellite peaks were not detected. Thus, again, LC-ECD provided a higher degree of structural information compared with nanoLC-EThcD.

To further validate the presence of a cysteine adduct, we repeated the analysis with the stronger reducing agent, TCEP,[46] at low and high concentration of urea. LC-ECD MS/MS was then performed with quadrupole isolation of the triply protonated precursor ion throughout the entire run. The total ion chromatograms from the corresponding TCEP sample preparation conditions as well as from the original DTT reduction are shown in Figure S7. Only the DTT condition resulted in the detection of the trisulfide-CG adduct, eluting at 21.3 min. An alternative explanation, however, would be that DTT favored alternative cysteine chemistry. For example, Burlingame and co-workers found a +209 Da carbamidomethyl-DTT cysteine adduct[47] in proteomic analysis on several different instruments. The three possible structures and monoisotopic masses for the three different isobaric peptide annotations are shown in Table 1.

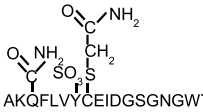
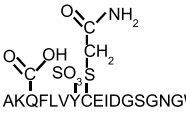
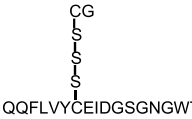
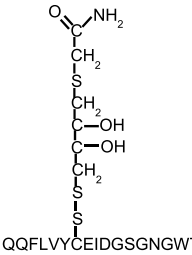
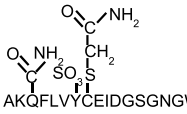
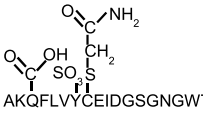
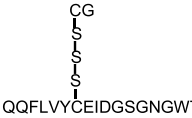
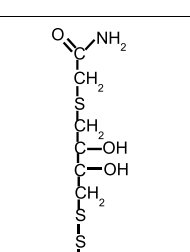
MS/MS of a Cysteinyl-Glycine Trisulfide vs Carbamidomethyl-Dithiothreitol Cysteine Adduct

Although the CAM-DTT adduct showed the lowest mass error in the MSFragger analysis of nanoLC-HCD MS/MS data, we noted that an increased concentration of DTT (5 mM compared with 200 μM) precluded detection of the +209 adduct. Thus, we generated both a CAM-DTT and trisulfide-CG adduct of the synthetic peptide QQVYCEIQK, a shorter version of the detected tryptic peptide **QQFLVY-CEIDGSGNGWTVFQK**. ECD MS/MS spectra of both cysteine adducts are shown in Figure 5. Similar to ECD of the detected tryptic peptide (Figure 4), both spectra show fragmentation behavior characteristic of peptides with S–S bonds. Note that the precursor ion mass is different by ∼1 Da because deamidation did not occur. However, both spectra contain a dominant 1138.57 monoisotopic peak corresponding to the unmodified peptide (R_1SH) generated by the S–S bond

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Table 1. Mass Shifts from the Observed Precursor Ion Mass for the MSFragger-Identified Sulfopeptide Candidate, AKQFLVYCEIDGSGNGWTVFQK, Compared to Its Sequence Variant, QQFLVYCEIDGSGNGWTVFQK, with Other Proposed Modifications

Sample	Suggested Peptide Modifications[a]	Observed Mass	Calibrated Observed Mass	Calculated Peptide Mass	Delta Mass	ppm error
Bovine fibrinogen digest	 AKQFLVYCEIDGSGNGWTVFQK	2627.148	2627.1482	2626.1827	0.9655	367.51
	 AKQFLVYCEIDGSGNGWTVFQK	2627.148	2627.1482	2627.1667	-0.0185	7.04
	 QQFLVYCEIDGSGNGWTVFQK	2627.148	2627.1482	2627.1125	0.0357	13.59
	 QQFLVYCEIDGSGNGWTVFQK	2627.148	2627.1482	2627.1489	-0.0007	0.27
Bovine plasma digest	 AKQFLVYCEIDGSGNGWTVFQK	2627.152	2627.1494	2626.1827	0.9667	367.97
	 AKQFLVYCEIDGSGNGWTVFQK	2627.152	2627.1494	2627.1667	-0.0173	6.59
	 QQFLVYCEIDGSGNGWTVFQK	2627.152	2627.1494	2627.1125	0.0369	14.05
	 QQFLVYCEIDGSGNGWTVFQK	2627.152	2627.1494	2627.1489	0.0005	0.19

[a]Tyrosine sulfation: +79.9568 Da, cysteine carbamidomethylation: +57.0215 Da, trisulfide-linked cysteinyl glycine: +207.997586 Da, glutamine deamidation: +0.984 Da, and cysteine carbamidomethylated dithiothreitol: +209.018024 Da.

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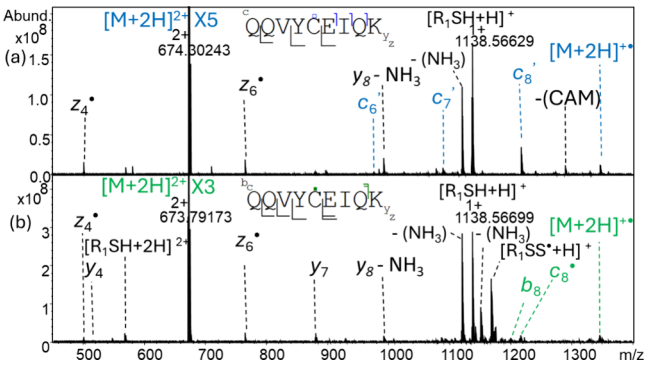


Figure 5. ECD MS/MS of synthetic peptide QQVYCEIQK (R_1) after incubation with DTT and IAA overnight to form a CAM-DTT cysteine adduct (a) and after disulfide bond formation with CG followed by conversion of the disulfide to a trisulfide bond (b). Structurally informative fragmentation is observed in each case. The blue open square indicates the CAM-DTT cysteine modification, whereas the green solid square denotes the trisulfide-linked CG. Fragments including the modifications are shown in the same colors.

cleavage. There are also some interesting differences between these spectra, including an abundant $R_1SS\bullet$ fragment for the trisulfide-containing peptide pair. This fragment results from cleavage of the second S−S bond. In addition, there are fewer backbone bond cleavages observed for the trisulfide-containing analyte, presumably because S−S bond cleavage is favored in ECD. Alternatively, the trisulfide-containing peptide contains an additional free amine compared with the CAM-DTT adducted peptide; i.e., additional charge isomers/gas-phase structures that affect ECD outcomes may exist. Furthermore, the CAM-DTT adduct shows CAM loss, which upon further inspection is also present in the previous data (Figure 4), marked by an asterisk. Furthermore, considering a CAM-DTT adduct rather than a trisulfide-CG adduct, additional *c*-type ions can be annotated in Figure 4 (Figure S8).

With further evidence that the potential sulfopeptide identified by MSFragger corresponds to a CAM-DTT adduct, we also re-examined the HCD and EThcD spectra. Considering this peptide structure, additional *b*-type ions could be identified (Figures S9 and S10). In addition, an abundant fragment at m/z 210.025, presumably corresponding to protonated CAM-DTT, is noted in both spectra. HCD MS^3 of this fragment showed an abundant MS^3 fragment at m/z 58.029, corresponding to protonated CAM (Figure S11). This ion also appears to be present for CAM-DTT-adducted peptides identified by Burlingame and co-workers. We propose that it can be used as a marker for CAM-DTT sample preparation artifacts.

CONCLUSIONS

LC-ECD and nanoLC EThcD MS/MS generated sulfate-retaining fragments, allowing for direct determination of tyrosine sulfation in a proteomic workflow. Although HCD leads to complete SO_3 neutral loss, it provides higher peptide sequence coverage than the radical-driven MS/MS methods. HCD data are also more compatible with various powerful bioinformatics search tools such as Proteome Discoverer and MSFragger. However, for maximum confidence and ability to discover novel modifications, the combination of multiple fragmentation techniques and multiple search engines with manual spectral interpretation along with gas-phase chemistry knowledge is crucial. In particular, we found that the accurate MS^1 mass shift analysis performed by MSFragger is valuable for differentiating sulfation from phosphorylation; however, such analysis can also lead to false sulfopeptide discoveries such as the discussed sample preparation artifact. Although, in our hands, LC-ECD provided superior structural information compared with nanoLC-EThcD, the latter MS/MS technique is more widely implemented. On the other hand, alternative ECD implementations are becoming more widely available.[48−50] Nevertheless, further work for enhancing sulfopeptide ETD efficiency, including implementation of supercharging agents or N-terminal guanidination,[15] will be needed.

ASSOCIATED CONTENT

Data Availability Statement

nanoLC-MS/MS (Orbitrap Fusion Lumos) and LC-MS/MS (SolariX FT-ICR) data files are publicly available through the ProteomeX-change Consortium via the PRIDE Repository with the data set identifier PXD048811.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jproteome.4c00072.

(Figure S1) NanoLC-HCD MS/MS analysis of bovine plasma tryptic digest on an Orbitrap Fusion Lumos; (Figure S2) NTL-HCD MS/MS analysis of bovine fibrinogen tryptic digest on an Orbitrap Fusion Lumos; (Figure S3) EThcD of fibrinogen sulfopeptides at various supplemental NCE; (Figure S4) LC-ECD MS/MS analysis of tryptically digested plasma performed on a 7 T SolariX FT-ICR; (Figure S5) total ion chromatogram from LC-ECD MS/MS time-segmented method; (Figure S6) EThcD spectrum of cysteinyl glycine trisulfide linked QQFLVYCEIDGSGNGWTVFQK peptide; (Figure S7) overlaid view of fibrinogen trypsin digested samples' total ion chromatograms with different reducing agents; (Figure S8) ECD spectrum of CAM-DTT cysteine modified QQFLVYCEIDGSGNGWTVFQK; (Figure S9) reannotated HCD spectrum of CAM-DTT cysteine modified QQFLVYCEIDGSGNGWTVFQK; (Figure S10) reannotated EThcD spectrum of CAM-DTT cysteine modified QQFLVYCEIDGSGNGWTVFQK; and (Figure S11) HCD MS^3 spectrum of the m/z 210.025 fragment from the CAM-DTT cysteine modified peptide (PDF)

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Funding

This work was supported by NIH grant R01GM139916 and an Agilent Thought Leader award to K.H. M.M.Y. acknowledges the financial support from the Ministry of Higher Education of the Arab Republic of Egypt. The Orbitrap Fusion Lumos was acquired through NIH grant S10OD021619.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

Joshua Salem is thanked for valuable discussions.

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https://doi.org/10.1021/acs.jproteome.4c00072
J. Proteome Res. 2024, 23, 2386−2396